Aryan Chahardovalee

Electrical Engineer | RF, EMC Test, Flight Test, Embedded & Autonomous Systems

Vancouver, BC · San Francisco Bay Area | +1 (415) 800-3435 | arianchahardovalee@gmail.com | aryanch.com | linkedin.com/in/aryan-chahardovalee
GitHub | Amateur Radio: VA7AYO

SUMMARY

Electrical engineer with 5+ years of hands-on experience across EMC testing, autonomous aircraft flight testing, UAS system validation, RF communications, embedded hardware, and test documentation. Experienced in laboratory measurement and troubleshooting, safety-critical test execution, regression analysis, and translating test results into clear engineering feedback.

SKILLS

EMC & RF Test: Radiated/conducted emissions and immunity, ESD/EFT/surge, chamber testing, RF troubleshooting, spectrum/network analyzers, VNAs, oscilloscopes, antenna characterization, S-parameters, SDR, BER and link budgets

Flight Test & Systems: Test planning, readiness, briefings/debriefings, procedures, UAS preflight/recovery, aircraft systems, communication reliability, environmental limitations, anomaly investigation, regression analysis

Hardware & Tools: PCB bring-up and rework, soldering, embedded systems, sensor integration, Python automation, SCPI/VISA, C/C++, MATLAB, Linux, Verilog, FEKO, Git/GitHub

EXPERIENCE

EMC Test Engineer - Cisco | Aug 2026 to Present

- Perform EMC laboratory testing on network hardware, including switches, routers, and access points, across emissions and immunity evaluations.
- Support chamber and bench testing, investigate device immunity issues, troubleshoot RF/electromagnetic failures, and document test setups and results.

Flight Test Engineer - Zipline International | Nov 2024 to Jun 2026

- Led flight-test validation within a small team, ensuring safety, procedural discipline, and operational readiness for autonomous aircraft systems.
- Documented test outcomes, identified regressions, communicated results, and supported troubleshooting through resolution; developed procedures and troubleshooting guides with multidisciplinary teams.
- Identified operating limitations including wind sensitivity, signal integrity, communication reliability, and environmental constraints; added flight representativeness labels and exclusion reasons to improve regression analysis.

AI/Hardware Integration Engineer - Huawei Canada | 2022 to 2023

- Built and tested drone WLAN links among aircraft, sensors, and an AI board, improving response speed by approximately 250%; integrated camera, IMU, LiDAR, sensor power, and real-time perception inputs.
- Recalibrated camera and IMU systems, replaced damaged components after test failures, and documented architecture and test observations.

Electronics Engineering Intern - Seaspan ULC | 2021 to 2022

- Produced shipboard communications reports, test agendas, block diagrams, and trade studies; performed RF characterization, antenna testing, and EMC/EMI analysis using FEKO and HPC resources.
- Built Python analysis and test automation that improved test efficiency by 30%; evaluated marine communication systems and recommended reliability improvements.

RF & Hardware Engineer - UBC ORBIT Satellite Design Team | 2019 to 2023

- Developed and tested CubeSat RF communication and antenna deployer PCBs; integrated radio hardware and supported system integration across communications, payload, power, and controls.

SELECTED PROJECTS

EMC / EMI Analysis of MPV: Assessed electromagnetic compatibility for a multi-purpose vessel and documented antenna separation recommendations; reported over 18 dB reduction in worst-case co-channel interference.

Antenna Deployer Module: Designed and validated a CubeSat antenna deployment subsystem through thermal-vacuum and communications testing.

Collaborative PIC Design: Developed a silicon photonic integrated circuit to meet design specifications within 3 nm.

WLAN for Huawei Drone: Improved command-link response and reduced dropout events through channel-selection and integration changes.

EDUCATION & CREDENTIALS

B.A.Sc., Electrical Engineering - University of British Columbia, 2018-2023 | **Airline and Flight Operations Coursework** - BCIT, 2024 (two terms)

EGBC Engineer-in-Training | FAA Part 107 Remote Pilot Certificate | ROC-A Aeronautical Radio Qualification | Amateur Radio VA7AYO | Private pilot training, 70+ hours Cessna 152 | Google Cybersecurity Certificate